Proof of the Transcendence of the Euler–Mascheroni Constant γ

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October 28, 2026

Abstract

Qi Jin introduced in his paper a method for proving the irrationality of numbers. Building upon this approach, we present an optimized version of the method and use it to prove that the Euler–Mascheroni constant γ is transcendental, and hence irrational. Furthermore, we establish the transcendence of πe, π/e, $\pi + e$ and $\pi - e$. γ, πe, π/e, $\pi + e$ and $\pi - e$ are both irrational and transcendental. In this paper, π denotes the Wallis product π_w, and γ denotes the Maclaurin series γ_M. Finally, it is proved that all $\zeta(k)$ are transcendental.

keywords: Euler–Mascheroni constant, γ, transcendence, irrationality, πe
MSC2020: 11J81

A preprint of this article is posted on Zenodo at
`https://doi.org/10.5281/zenodo.23282062`.

1 Introduction

Qi Jin introduced in his paper a method for proving the irrationality of numbers [1].Building upon this approach, we present an optimized version of the method and use it to prove that the Euler–Mascheroni constant γ is transcendental, and hence irrational. Furthermore, we establish the transcendence of πe, π/e, $\pi + e$ and $\pi - e$. Qi Jin's proof is based on the definition of irrational numbers, whereas this paper proceeds from the definition of transcendental numbers.

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Email: 871685523@qq.com. No funding was received for this work. No part of this article was generated by AI.

2 Concept

[1, p. 6] The set of natural numbers is divided into two parts: one part consists of those that can be expressed by numerical expressions, referred to as expressible natural numbers; the other part consists of those that cannot be expressed by numerical expressions, referred to as inexpressible natural numbers, as shown below:

$$\underbrace{0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad \cdots}_{\text{expressible natural numbers}} \underbrace{N-3 \quad N-2 \quad N-1 \quad N}_{\text{inexpressible natural numbers}}$$

The upper bound of the natural numbers is N. Any number exceeding N, such as $2N$ or N^2, is no longer a natural number.

All expressible natural numbers are finite. The sum or product of two finite natural numbers remains finite. Moreover, the sum or product of finitely many finite natural numbers is also finite; that is, if a and b are finite, then $a^b < N$.

3 Lemma 1. An Inequality Concerning $f(N)$

Let N be the largest natural number, and let $f(N)$ be a function of N. If $f(N)$ exceeds every finite natural number (i.e., $f(N) \to \infty$), denoting by k_i an arbitrary finite natural number for each $i = 1, 2, \ldots, N$, we obtain the following relation:

$$\begin{aligned}
& f(N) > k_{i-1} \\
\Rightarrow\ & f(N) > k_{i-1} + k_i \\
\Rightarrow\ & f(N) - k_i > k_{i-1} \\
\Rightarrow\ & f(N) - k_i + (i-1)f(N) > k_{i-1} + (i-1)f(N) \\
\Rightarrow\ & if(N) - k_i > (i-1)f(N) + k_{i-1} \\
\Rightarrow\ & if(N) + k_i > if(N) - k_i > (i-1)f(N) + k_{i-1} > (i-1)f(N) - k_{i-1} \\
\Rightarrow\ & if(N) \pm k_i > (i-1)f(N) \pm k_i.
\end{aligned}$$

Substituting $i = 1, 2, 3, \ldots, N$ yields the following inequality concerning $f(N)$:

$$Nf(N) \pm k_N > (N-1)f(N) \pm k_{N-1} > \cdots > 2f(N) \pm k_2 > f(N) \pm k_1 > \pm k_0.$$

4 Theorem 1. A Sufficient Condition for Transcendental Numbers

Let the canonical prime factorization of a positive integer n be $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$. Define $\omega(n) = k$ as the number of distinct prime factors of n, and $\Omega(n) = \alpha_1 + \alpha_2 + \cdots + \alpha_k$ as the total number of prime factors of n counted with multiplicity.

Let P be an arbitrary prime and $f(N)$ a function of N. If a number S can be expressed as the following irreducible fraction:

$$S = \frac{B}{A} = \frac{P^{f(N)}E}{D}, \quad \gcd(P, E) = 1,$$

or

$$S = \frac{B}{A} = \frac{E}{P^{f(N)}D}, \quad \gcd(P, D) = 1,$$

where $A \neq 0$, $B \neq 0$, $E \neq 0$, and $D \neq 0$, then $\Omega(A) \to \infty$ or $\Omega(B) \to \infty$ is a sufficient condition for S to be transcendental. A necessary and sufficient condition for $\Omega(A) \to \infty$ or $\Omega(B) \to \infty$ is that $f(N) \to \infty$ or $\omega(A) \to \infty$ or $\omega(B) \to \infty$.

We now proceed to prove this proposition.

4.1 $f(N) \to \infty$

For any monomial polynomial $F(x) = ax^k$ with $a \neq 0$, we have $F(S) \neq 0$. Henceforth, we only consider polynomials with more than one term; that is, polynomials consisting of at least two terms.

First, consider the case where $S = \dfrac{P^{f(N)}E}{D}$, i.e., when $P^{f(N)}$ appears in the numerator (the $\pm$ signs of each term are chosen independently; the choice of sign at any position is independent of those at other positions):

$$F(x) = \pm a_n x^n \pm a_{n-1} x^{n-1} \pm \cdots \pm a_1 x \pm a_0.$$

Express the coefficients a_i as irreducible fractions:

$$a_i = \frac{P^{c_i} g_i}{b_i} \qquad \text{or} \qquad a_i = \frac{g_i}{P^{c_i} b_i},$$

where all b_i, g_i, c_i, n are finite natural numbers, $b_i \neq 0$, $\gcd(P, g_i) = 1$, and $\gcd(P, b_i) = 1$.

Substituting S into x^i yields:

$$x^i = S^i = \frac{P^{if(N)}E^i}{D^i}.$$

When P^{c_0} appears in the numerator, i.e., $a_0 = \dfrac{P^{c_0}g_0}{b_0}$, adopting the conventions that $0 \cdot f(n) = 0$ and $x^0 = 1$, we have:

$$a_i x^i = \frac{P^{if(N)\pm c_i}E^i g_i}{D^i b_i};$$

$$F(S) = \sum_{i=1}^{n} \pm \frac{P^{if(N)\pm c_i}E^i g_i}{D^i b_i} \pm \frac{P^{c_0}g_0}{b_0}.$$

After combining the terms over a common denominator and simplifying the numerator of each term $a_i x^i$ to the form $P^{if(N)\pm c_i}G_i$, where $\gcd(P, G_i) = 1$, we obtain:

$$F(S) = \sum_{i=1}^{n} \pm \frac{P^{if(N)\pm c_i}G_i}{D^n \prod b_i} \pm \frac{P^{c_0}G_0}{D^n \prod b_i}.$$

Whether $F(S)$ vanishes depends entirely on the numerator; the denominator need not be considered at all.

The inequality concerning $f(N)$ from **Lemma 1**, namely

$$Nf(N) \pm k_N > (N-1)f(N) \pm k_{N-1} > \cdots > 2f(N) \pm k_2 > f(N) \pm k_1 > \pm k_0,$$

ensures that the powers of P in any two terms of the numerator are distinct. Arbitrarily partition these terms into two groups, A and B, each containing at least one term. Let L_A denote the minimum exponent of P among all terms in group A, and let L_B denote the minimum exponent of P among all terms in group B. Then it necessarily follows that:

$$\mathbf{L_A \neq L_B}.$$

Let $\mathrm{SUM}(A)$ denote the sum of the numerators of all terms in group A. Then we have:

$$\mathrm{SUM}(A) = \sum_{i \in A} \pm P^{L_A} H_{Ai} = P^{L_A} \sum_{i \in A} \pm H_{Ai}, \quad \gcd\left(P, \sum_{i \in A} \pm H_{Ai}\right) = 1.$$

Let $\mathrm{SUM}(B)$ denote the sum of the numerators of all terms in group B. Then we have:

$$\mathrm{SUM}(B) = \sum_{i\in B} \pm P^{L_B} H_{Bi} = P^{L_B} \sum_{i\in B} \pm H_{Bi}, \quad \gcd\left(P, \sum_{i\in B} \pm H_{Bi}\right) = 1.$$

By the Fundamental Theorem of Arithmetic, if two numbers have equal absolute values, their factorizations must be identical up to sign. Thus:

$$\begin{aligned} L_A \neq L_B &\Rightarrow |\mathrm{SUM}(A)| \neq |\mathrm{SUM}(B)| \\ &\Rightarrow \mathrm{SUM}(A) \pm \mathrm{SUM}(B) \neq 0 \\ &\Rightarrow \mathbf{F}(\mathbf{S}) \neq \mathbf{0}. \end{aligned}$$

This implies that S cannot be a root of any algebraic equation; hence, S is transcendental and, consequently, irrational.

When P^{c_0} appears in the denominator, i.e., $a_0 = \dfrac{g_0}{P^{c_0} b_0}$, we have:

$$F(S) = \sum_{i=1}^{n} \pm \frac{P^{if(N)\pm c_i} E^i g_i}{D^i b_i} \pm \frac{g_0}{P^{c_0} b_0},$$

$$P^{c_0} F(S) = \sum_{i=1}^{n} \pm \frac{P^{if(N)\pm c_i + c_0} E^i g_i}{D^i b_i} \pm \frac{g_0}{b_0}.$$

By the same method as above, we can prove the following:

$$P^{c_0} F(S) \neq 0 \Rightarrow F(S) \neq 0.$$

This implies that S cannot be a root of any algebraic equation; hence, S is transcendental and, consequently, irrational.

When $S = \dfrac{E}{P^{f(N)} D}$, i.e., when $P^{f(N)}$ appears in the denominator, the argument is simpler since the term a_0 need not be considered. We have:

$$F(S) = \sum_{i=1}^{n} \pm \frac{E^i g_i}{P^{if(N)\mp c_i} D^i b_i} \pm P^{\pm c_0} \frac{g_0}{b_0}.$$

After combining the terms over the common denominator $P^{nf(N)\mp c_n} D^n \prod b_i$ and simplifying the numerator of each term $a_i x^i$ to the form $P^{(n-i)f(N)\mp c_n \pm c_i} G_i$, where $\gcd(P, G_i) = 1$, we obtain:

$$F(S) = \sum_{i=1}^{n} \pm \frac{P^{(n-i)f(N)\mp c_n \pm c_i} G_i}{P^{nf(N)\mp c_n} D^n \prod b_i} \pm \frac{P^{nf(N)\mp c_n \pm c_0} G_0}{P^{nf(N)\mp c_n} D^n \prod b_i}.$$

By the same reasoning, it follows that $F(S) \neq 0$. Hence, S must be transcendental.

This proves that $f(N) \to \infty$ is a sufficient condition for S to be transcendental.

4.2 $\omega(A) \to \infty$ or $\omega(B) \to \infty$

$\omega(B) \to \infty$ implies that the numerator B of $S = B/A$ has infinitely many distinct prime factors. Since the numerators and denominators of the polynomial's coefficients contain only finitely many prime factors, there must exist a prime factor T of B that does not divide the numerator or denominator of any coefficient of the polynomial. In other words, the exponent of T is unaffected by the coefficients and depends solely on x^i. Let the power of T be k, where k is a finite number. The term x^i will similarly yield an inequality analogous to that in Lemma 1:

$$T^{Nk} > T^{(N-1)k} > \cdots > T^{3k} > T^{2k} > T^k > 1.$$

After finding a common denominator and summing the polynomials, the resulting numerator takes the following form:

$$F(T) = Z_n T^{nk} \pm Z_{n-1} T^{(n-1)k} \pm \cdots \pm Z_2 T^{2k} \pm Z_1 T^k \pm Z_0,$$

Since the Z_i are integers with $\gcd(Z_i, T) = 1$, the same argument as above shows that $F(T) \neq 0$.

As $\omega(A) \to \infty$, since T appears in the denominator, combining terms over the common denominator T^{nk} yields the inequality:

$$T^{(n-n)k} < T^{nk-(n-1)k} < \cdots < T^{(n-2)k} < T^{(n-1)k} < T^{(n-0)k},$$

that is,

$$1 < T^k < T^{2k} < \cdots < T^{(n-2)k} < T^{(n-1)k} < T^{nk}.$$

which similarly implies $F(T) \neq 0$, and thus S is transcendental. Therefore, $\omega(A) \to \infty$ or $\omega(B) \to \infty$ is also a sufficient condition for S to be transcendental.

This establishes that $\Omega(A) \to \infty$ or $\Omega(B) \to \infty$ is a sufficient condition for the transcendence of S, thereby completing the proof of the theorem.

5 γ Is Transcendental and Irrational

From the definition of the Euler–Mascheroni constant, we have:

$$\gamma = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln n \right) = \sum_{k=1}^{N} \frac{1}{k} - \ln N$$

$$= \sum_{k=1}^{N} \frac{1}{k} - \ln\left(\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{N-1}{N-2} \cdot \frac{N}{N-1}\right)$$

$$= \sum_{k=1}^{N} \frac{1}{k} - \sum_{k=1}^{N-1} \ln \frac{k+1}{k} = \sum_{k=1}^{N} \frac{1}{k} - \sum_{k=1}^{N-1} \ln\left(1 + \frac{1}{k}\right).$$

The Taylor expansion of $\ln(1+x)$ about $x = 0$ is given by:

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{x^N}{N}, \qquad -1 < x \leq 1.$$

Substituting the respective values yields (note that N may be either odd or even, and consecutive terms alternate in sign):

$$\ln \frac{2}{1} = \ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \pm \frac{1}{N-1} \mp \frac{1}{N};$$

$$\ln \frac{3}{2} = \ln\left(1 + \frac{1}{2}\right) = \frac{1}{2} - \frac{1}{2^2 \cdot 2} + \frac{1}{2^3 \cdot 3} - \frac{1}{2^4 \cdot 4} + \cdots \pm \frac{1}{2^{N-1}(N-1)} \mp \frac{1}{2^N N};$$

$$\ln \frac{4}{3} = \ln\left(1 + \frac{1}{3}\right) = \frac{1}{3} - \frac{1}{3^2 \cdot 2} + \frac{1}{3^3 \cdot 3} - \frac{1}{3^4 \cdot 4} + \cdots \pm \frac{1}{3^{N-1}(N-1)} \mp \frac{1}{3^N N};$$

$$\ln \frac{5}{4} = \ln\left(1 + \frac{1}{4}\right) = \frac{1}{4} - \frac{1}{4^2 \cdot 2} + \frac{1}{4^3 \cdot 3} - \frac{1}{4^4 \cdot 4} + \cdots \pm \frac{1}{4^{N-1}(N-1)} \mp \frac{1}{4^N N};$$

$$\vdots$$

$$\ln \frac{N}{N-1} = \ln\left(1 + \frac{1}{N-1}\right)$$
$$= \frac{1}{N-1} - \frac{1}{(N-1)^2 \cdot 2} + \frac{1}{(N-1)^3 \cdot 3} - \frac{1}{(N-1)^4 \cdot 4} + \cdots$$
$$\pm \frac{1}{(N-1)^{N-1}(N-1)} \mp \frac{1}{(N-1)^N N}.$$

Among the natural numbers up to N, there necessarily exists a power of two, 2^h, satisfying

$$\frac{N}{2} < 2^h \leq N.$$

This 2^h is the unique integer in $\{1, 2, \ldots, N\}$ with the highest power of 2 in its factorization; for every other natural number in this range, the exponent of 2 in its

prime factorization is at most $h-1$. We have:

$$\ln\left(1+\frac{1}{2^h}\right) = \frac{1}{2^h} - \frac{1}{2^{2h}\cdot 2} + \frac{1}{2^{3h}\cdot 3} - \frac{1}{2^{4h}\cdot 4} + \cdots \\ \pm \frac{1}{2^{(N-3)h}(N-3)} \mp \frac{1}{2^{(N-2)h}(N-2)} \pm \frac{1}{2^{(N-1)h}(N-1)} \mp \frac{1}{2^{Nh}N}.$$

Let $v_p(n)$ denote the p-adic valuation of n (i.e., the exact exponent of the prime p in the prime factorization of n). When $N-1 \neq 2^h$, among the denominators of all terms, the one with the highest power of 2 is $2^{Nh}N$. We have:

$$v_2(2^{Nh}N) = Nh + v_2(N).$$

Let $v_2(\text{other})$ denote the highest power of 2 in the denominators of the remaining terms. Then we have:

$$v_2(\text{other}) < Nh + v_2(N).$$

When $N-1 = 2^h$, since $\gcd(N, N-1) = 1$ and $v_2(N) = 0$, we have:

$$v_2(2^{(N-1)h}(N-1)) = v_2(2^{Nh}N) = Nh.$$

Combining the last two terms of this expansion yields:

$$\frac{1}{2^{(N-1)h}(N-1)} - \frac{1}{2^{Nh}N} = \frac{1}{2^{(N-1)h}2^h} - \frac{1}{2^{Nh}(2^h+1)}$$

$$= \frac{1}{2^{Nh}} - \frac{1}{2^{Nh}(2^h+1)} = \frac{2^h}{2^{Nh}(2^h+1)} = \frac{1}{2^{(N-1)h}(2^h+1)}.$$

After combining the last two terms of the expansion of $\ln(1+1/2^h)$, the term whose denominator contains the highest power of 2 among all terms is precisely this combined term, with denominator $2^{(N-1)h}(2^h+1)$. We have:

$$v_2(2^{(N-1)h}(2^h+1)) = (N-1)h,$$

$$v_2(\text{other}) < (N-1)h.$$

Rewrite each term in the form $\dfrac{1}{2^j X_{il}}$, where $\gcd(2, X_{il}) = 1$. There exists a unique term for which j attains its maximum value; denote this maximal exponent by $j_{\max}$ and the corresponding X_{il} by $X_{\max}$. For all remaining terms, denote the exponent by j_{other} and the corresponding odd factor by X_{other}, where $j_{\text{other}} \leq j_{\max} - 1$.

Taking $(N!)^N$ as the common denominator to combine the sum, and writing $(N!)^N = 2^T W$, with $\gcd(2, W) = 1$, we obtain:

$$\gamma = \sum \frac{1}{2^j X_{il}} = \frac{\sum \frac{2^T W}{2^j X_{il}}}{2^T W} = \frac{\sum \frac{2^{T-j} W}{X_{il}}}{2^T W}$$

$$= \frac{\frac{2^{T-j_{\max}} W}{X_{\max}} + \sum \frac{2^{T-j_{\text{other}}} W}{X_{\text{other}}}}{2^T W}$$

$$= \frac{2^{T-j_{\max}} \left(\frac{W}{X_{\max}} + \sum \frac{2^{j_{\max}-j_{\text{other}}} W}{X_{\text{other}}} \right)}{2^T W}$$

$$= \frac{\frac{W}{X_{\max}} + \sum \frac{2^{j_{\max}-j_{\text{other}}} W}{X_{\text{other}}}}{2^{j_{\max}} W}$$

$$= \frac{\frac{W}{X_{\max}} + 2 \sum \frac{2^{j_{\max}-j_{\text{other}}-1} W}{X_{\text{other}}}}{2^{j_{\max}} W}.$$

Since $j_{\max} - j_{\text{other}} - 1 \geq 0$, we have:

$$\gcd\left(2,\, 2 \sum \frac{2^{j_{\max}-j_{\text{other}}-1} W}{X_{\text{other}}}\right) = 0.$$

As $\gcd(2, W/X_{\max}) = 1$, it follows that:

$$\gcd\left(2,\, \frac{W}{X_{\max}} + 2 \sum \frac{2^{j_{\max}-j_{\text{other}}-1} W}{X_{\text{other}}}\right) = 1.$$

Consequently, the factor $2^{j_{\max}}$ in the denominator of γ cannot be canceled.

Since $j_{\max} = Nh + v_2(N)$ or $(N-1)h$, we have $f(N) \geq (N-1)h \to \infty$, which satisfies the sufficient condition of **Theorem 1**. Therefore, γ is both transcendental and irrational.

6 π/e and πe Are Transcendental and Irrational

The natural number N may be either odd ($N = 2M+1$) or even ($N = 2M+2$).

According to the Wallis Product, we have:

$$\pi = 2\prod_{n=1}^{\infty}\frac{4n^2}{4n^2-1} = 2\prod_{n=1}^{M}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right).$$

When $N = 2M+1$, based on the definition of e, we have:

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \left(1+\frac{1}{2M}\right)^{2M} = \left(\frac{2M+1}{2M}\right)^{2M}.$$

$$\begin{aligned}\frac{\pi}{e} &= \left[2\prod_{n=1}^{M}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right)\right]\frac{(2M)^{2M}}{(2M+1)^{2M}} \\ &= \frac{2^{2M+1}\,M!\,M!\,(2M)^{2M}}{(2M-1)!!\,(2M+1)!!\,(2M+1)^{2M}}.\end{aligned}$$

After simplification, all factors in the denominator are odd, while the factor 2^{2M+1} in the numerator satisfies $f(N) = 2M+1 = N \to \infty$. According to Theorem 1, π/e must be transcendental and irrational.

When $N = 2M+2$, based on the definition of e, we have:

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \left(1+\frac{1}{2M+1}\right)^{2M+1} = \left(\frac{2M+2}{2M+1}\right)^{2M+1}.$$

$$\begin{aligned}\frac{\pi}{e} &= \left[2\prod_{n=1}^{M}\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right]\frac{(2M+1)^{2M+1}}{(2M+2)^{2M+1}} \\ &= \frac{2^{2M+1}\,M!\,M!\,(2M+1)^{2M+1}}{(2M-1)!!\,(2M+1)!!\,(2M+2)^{2M+1}}.\end{aligned}$$

Since $\gcd(2M+1,\, 2M+2) = 1$, the term $(2M+1)$ cannot be canceled with $(2M+2)$. Considering the following fraction alone:

$$\frac{(2M+1)^{2M+1}}{(2M-1)!!\,(2M+1)!!} = \frac{(2M+1)^{2M}}{(2M-1)!!\,(2M-1)!!}.$$

When $2M+1$ is prime, there is no factor in the denominator that can cancel it; hence

$$f(N) = 2M = N-2 \to \infty,$$

which satisfies the sufficient condition in Theorem 1.

When $2M+1$ has exactly one prime factor p, we have $v_p(2M+1) = \log_p(2M+1)$, and thus:

$$v_p\big((2M+1)^{2M}\big) = \big[\log_p(2M+1)\big]^{2M}.$$

It follows that the power of p in the numerator far exceeds that in the denominator; namely:

$$f(N) = \big[\log_p(2M+1)\big]^{2M} - \frac{2(2M-1)}{p-1} \to \infty,$$

which also satisfies the sufficient condition in Theorem 1.

When the odd number $M+1$ has multiple prime factors, at least one prime factor p satisfies $p \geq 5$. By Legendre's formula, it is well known that

$$v_p(n!) = \frac{n}{p-1} + O(\log n),$$

and in particular $v_p(n!) \sim \dfrac{n}{p-1}$ as $n \to \infty$. Hence we have:

$$v_p\big((2M-1)!!\,(2M-1)!!\big) \leq v_p\big((2M-1)!\,(2M-1)!\big) \sim \frac{2(2M-1)}{p-1}.$$

Comparing the powers of p in the numerator and denominator:

$$v_p\big((2M+1)^{2M}\big) \geq 2M;$$

$$v_p\big((2M-1)!!\,(2M-1)!!\big) \leq v_5\big((2M-1)!!\,(2M-1)!!\big)$$

$$\leq v_5\big((2M-1)!\,(2M-1)!\big) \sim \frac{2(2M-1)}{5-1} = \frac{2M-1}{2};$$

we observe that at least a factor of

$$p^{\,2M-(2M-1)/2} = p^{\,M+1/2}$$

in the numerator remains uncanceled. Hence $f(N) \geq M + \frac{1}{2} \to \infty$, which satisfies the sufficient condition of Theorem 1.

Therefore π/e is transcendental and irrational. Likewise, πe can be shown to be transcendental and irrational.

7 $\pi + e$ and $\pi - e$ are Transcendental and Irrational

When $N = 2M + 1$, based on the definition of e, we have:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = \left(\frac{2M+1}{2M}\right)^{2M} = \frac{(2M+1)^{2M}}{2^{2M} M^{2M}}.$$

$$\pi = \frac{2^{2M+1} M!M!}{(2M-1)!!(2M+1)!!}.$$

$$\begin{aligned}\pi + e &= \frac{2^{2M+1} M!M!}{(2M-1)!!(2M+1)!!} + \frac{(2M+1)^{2M}}{2^{2M} M^{2M}} \\ &= \frac{2^{2M+1} M!M! \cdot 2^{2M} M^{2M} + (2M-1)!!(2M+1)!!(2M+1)^{2M}}{(2M-1)!!(2M+1)!! \cdot 2^{2M} M^{2M}}.\end{aligned}$$

The numerator has no factor of 2, so the 2^{2M} in the denominator cannot be canceled; hence, $\pi + e$ is transcendental and irrational.

When $N = 2M + 2$, based on the definition of e, we have:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = \left(\frac{2M+2}{2M+1}\right)^{2M+1} = \frac{\left[2^{2M+1}(M+1)\right]^{2M+1}}{(2M+1)^{2M+1}}.$$

$$\begin{aligned}\pi + e &= \frac{2^{2M+1} M!M!}{(2M-1)!!(2M+1)!!} + \frac{\left[2^{2M+1}(M+1)\right]^{2M+1}}{(2M+1)^{2M+1}} \\ &= \frac{2^{2M+1}\left[M!M!(2M+1)^{2M+1} + (M+1)^{2M+1}(2M-1)!!(2M+1)!!\right]}{(2M-1)!!(2M+1)!!(2M+1)^{2M+1}}.\end{aligned}$$

The denominator has no factor of 2, so the 2^{2M} in the numerator cannot be canceled; hence, $\pi + e$ is transcendental and irrational.

The same method can be used to prove that $\pi - e$ is transcendental and irrational.

8 Transcendence of $\zeta(k)$

The primitive definition of $\zeta(k)$ is the value of the Riemann zeta function at positive integers. The method used to prove the transcendence of γ can similarly be applied to prove that all $\zeta(k)$ are transcendental numbers.

$$\zeta(k) = \sum_{n=1}^{\infty} \frac{1}{n^k}, \qquad k > 1.$$

Specifically,

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{N^3},$$
$$\zeta(7) = \sum_{n=1}^{\infty} \frac{1}{n^7} = \frac{1}{1^7} + \frac{1}{2^7} + \frac{1}{3^7} + \cdots + \frac{1}{N^7},$$
$$\zeta(m) = \sum_{n=1}^{\infty} \frac{1}{n^m} = \frac{1}{1^m} + \frac{1}{2^m} + \frac{1}{3^m} + \cdots + \frac{1}{N^m}.$$

By using $(N!)^m$ as a common denominator, and writing $(N!)^m = 2^T W$, with $\gcd(2, W) = 1$, we obtain:

$$\begin{aligned}\zeta(m) &= \frac{\frac{(N!)^m}{1^m} + \frac{(N!)^m}{2^m} + \cdots + \frac{(N!)^m}{(N-1)^m} + \frac{(N!)^m}{N^m}}{(N!)^m} \\ &= \frac{\frac{2^T W}{1^m} + \frac{2^T W}{2^m} + \cdots + \frac{2^T W}{(N-1)^m} + \frac{2^T W}{N^m}}{2^T W}.\end{aligned}$$

Among the natural numbers up to N, there necessarily exists a power of two, 2^h, satisfying

$$\frac{N}{2} < 2^h \leq N.$$

This 2^h is the unique integer in $\{1, 2, \ldots, N\}$ with the highest power of 2 in its factorization; for every other natural number in this range, the exponent of 2 in its prime factorization is at most $h - 1$. Let the largest such exponent be denoted by $h_{\max}$, and the powers of 2 in the other terms be denoted by h_{other}. Then we have:

$$h_{\max} - h_{\text{other}} \geq 1.$$

Hence $\zeta(m)$ can be expressed as:

$$\begin{aligned}\zeta(m) &= \frac{2^{T - m h_{\max}} W + \sum \frac{2^{T - m h_{\text{other}}} W}{X_{\text{other}}}}{2^T W} \\ &= \frac{2^{T - m h_{\max}} \left(W + 2^{m(h_{\max} - h_{\text{other}})} \sum \frac{W}{X_{\text{other}}} \right)}{2^T W} \\ &= \frac{W + 2^{h_{\max} - h_{\text{other}}} \sum \frac{W}{X_{\text{other}}}}{2^{m h_{\max}} W}.\end{aligned}$$

The numerator is odd, and the factor $2^{mh_{\max}}$ in the denominator cannot be cancelled out. Since $N/2 < 2^{h_{\max}} \leq N$, it follows that:

$$h_{\max} > \log_2\left(\frac{N}{2}\right) \quad \Longrightarrow \quad mh_{\max} > m\log_2\left(\frac{N}{2}\right) \to \infty.$$

Therefore, all $\zeta(k)$ are transcendental numbers, and consequently, irrational numbers.

9 Classification of Numbers

9.1 Maclaurin Series of e

Define a function $\mathrm{trans}(\tau)$ concerning transcendental numbers as follows:

$$\mathrm{trans}(\tau) = \begin{cases} 1, & \text{if } \tau \text{ is transcendental,} \\ 0, & \text{if } \tau \text{ is algebraic.} \end{cases}$$

By the definition of e, it follows that $\mathrm{trans}(e) = 1$. Let e_M denote the Maclaurin series of e, namely:

$$e_M = \sum_{m=0}^{N} \frac{1}{m!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{(N-1)!} + \frac{1}{N!}.$$

It can be shown that e_M is transcendental, i.e., $\mathrm{trans}(e_M) = 1$. Taking $N!$ as the common denominator yields:

$$e_M = \frac{2N! + \frac{N!}{2!} + \cdots + N(N-1) + N + 1}{N!} = \frac{H}{N!}.$$

Since $N \mid (H-1)$, it follows that $\gcd(N, H) = 1$. Performing prime factorization on the number N, since $N \to \infty$, there exist only the following three cases:

Case 1: The exponent of some factor P tends to infinity, i.e., $N = P^z X$, where $z \to \infty$. Then e_M is transcendental.

Case 2: N has infinitely many prime factors. According to Theorem 1, e_M is transcendental.

Case 3: The factors include inexpressible infinite numbers. When N is prime, or when its prime factors include an inexpressible number—for example, $N = 3 \cdot \frac{N}{3}$, where $\frac{N}{3}$ is prime—let P denote this inexpressible prime factor. Then the factor P does not appear in either the numerator or the denominator of any coefficient a_i of the polynomial. As in Case 2, it follows that e_M is transcendental.

Although $e = e_M$ and $\text{trans}(e) = \text{trans}(e_M)$, e and e_M are two distinct numbers. It can be proved that their irreducible fractions contain different prime factors, or the exponents of the same factors are unequal. The proof is omitted.

9.2 Wallis Product for π

Let π_w denote the Wallis product for π, we have:

$$\pi_w = 2\prod_{n=1}^{\infty}\frac{4n^2}{4n^2-1} = 2\prod_{n=1}^{M}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right).$$

Since $\pi = \pi_w$ and $\text{trans}(\pi) = \text{trans}(\pi_w) = 1$, the Wallis product π_w may serve as a substitute for π in the study of its transcendence, in the absence of an algebraic definition of π.

9.3 Extension of the Definition of γ

The term $\ln(N)$ appearing in the definition of the Euler–Mascheroni constant γ is not directly amenable to analysis; we therefore adopt its Maclaurin series expansion γ_M as an alternative definition:

$$\begin{aligned}
\gamma_M &= \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \ln n\right) = \sum_{k=1}^{N}\frac{1}{k} - \ln N \\
&= \sum_{k=1}^{N}\frac{1}{k} - \ln\left(\frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{N-1}{N-2}\cdot\frac{N}{N-1}\right) \\
&= \sum_{k=1}^{N}\frac{1}{k} - \sum_{k=1}^{N-1}\ln\frac{k+1}{k} = \sum_{k=1}^{N}\frac{1}{k} - \sum_{k=1}^{N-1}\ln\left(1+\frac{1}{k}\right) \\
&= \sum_{k=1}^{N}\frac{1}{k} - \sum_{k=1}^{N-1}\sum_{m=1}^{N}\frac{(-1)^{m-1}}{m\,k^m}.
\end{aligned}$$

By Lindemann's theorem [2], e^{α} is transcendental for any algebraic number α. Consequently, $\ln\!\left(1+\frac{1}{k}\right)$ is necessarily transcendental, and the transcendence of

$$\sum_{m=1}^{N} \frac{(-1)^{m-1}}{mk^m}$$

can also be established via the same argument given above. Hence this extension of the definition is valid. Therefore this series representation γ_M may then be employed in the study of transcendence questions concerning γ.

9.4 Transcendental Approximation of Algebraic Numbers

It is well known that $\sqrt{2}$ is algebraic, i.e., $\mathrm{trans}(\sqrt{2}) = 0$. Algebraic numbers can be approximated by series of transcendental numbers. Let

$$\theta = 1 + \sum_{m=1}^{N} \frac{(-1)^{m-1}(2m-2)!}{2^{2m-1}\, m!\, (m-1)!}.$$

Since θ is precisely the Taylor series of $\sqrt{2}$, we have $\theta = \sqrt{2}$. However, Theorem 1 implies that θ is transcendental; that is,

$$\mathrm{trans}(\theta) = 1 \neq \mathrm{trans}(\sqrt{2}) = 0.$$

Therefore θ and $\sqrt{2}$ are two distinct numbers. Consequently, θ cannot be substituted for $\sqrt{2}$ when investigating the transcendence of $\sqrt{2}$.

In general, the transcendence of any given number must be studied solely from its definition. Arbitrary replacement by a series representation is not permissible.

Final Remark

In this paper, π denotes the Wallis product π_w, and γ denotes the Maclaurin series γ_M. The author does not establish whether π coincides with π_w, or whether γ coincides with γ_M. The results actually proved in this paper are that γ_M is transcendental, and that $\pi_w e$, π_w/e, and $\pi_w \pm e$ are all transcendental numbers.

References

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[2] Ferdinand von Lindemann, Ueber die Zahl π, *Math. Ann.* **20** (1882), 213–225. https://doi.org/10.1007/BF01446522